

```

import numpy as np

import pandas as pd

data = {
    'restaurant': ['A', 'B', 'A', 'C', 'B'],
    'delivery_time': [30, 45, np.nan, 25, 60],
    'expected_time': [25, 40, 30, 20, 50],
    'price': [200, -50, 150, 180, 0],
    'rating': [4, 5, 3, 4, 2]
}

df = pd.DataFrame(data) df =
df.drop_duplicates()

delayed = df[df['delivery_time'] > df['expected_time']] print(delayed)
  restaurant delivery_time expected_time price rating 0      A      30.0      25
200      4
1      B      45.0      40    -50      5
3      C      25.0      20    180      4
4      B      60.0      50      0      2
df['delivery_time'] =
df['delivery_time'].fillna(df['delivery_time'].mean())

df['price'] = np.where(df['price'] <= 0, df['price'].mean(), df['price']) print(df)
  restaurant delivery_time expected_time price rating 0      A      30.0      25
200.0      4
1      B      45.0      40    96.0      5
2      A      40.0      30    150.0      3
3      C      25.0      20    180.0      4
4      B      60.0      50    96.0      2
import plotly.express as px

fig1 = px.histogram(df, x='delivery_time', title="Delivery Time") fig1.show()

top = df['restaurant'].value_counts().reset_index()

```

```
top.columns = ['restaurant', 'count']
```

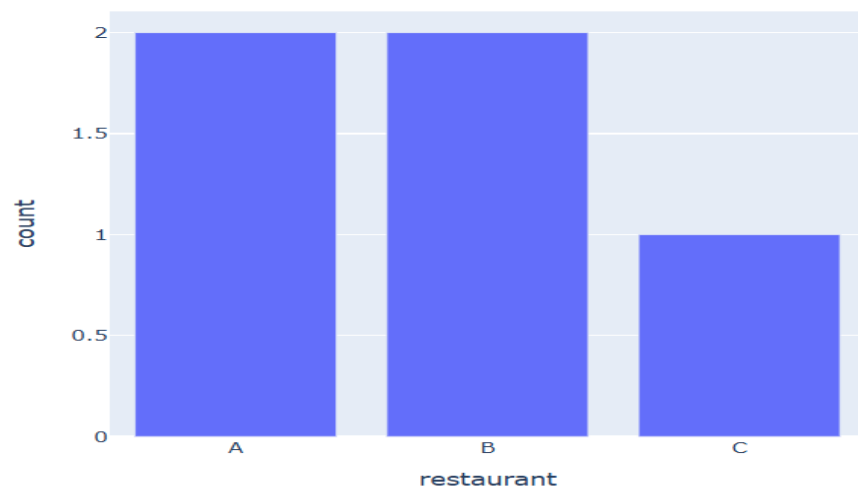
```
fig2 = px.bar(top, x='restaurant', y='count', title="Top Restaurants") fig2.show()
```

```
fig3 = px.box(df, y='rating', title="Ratings") fig3.show()
```

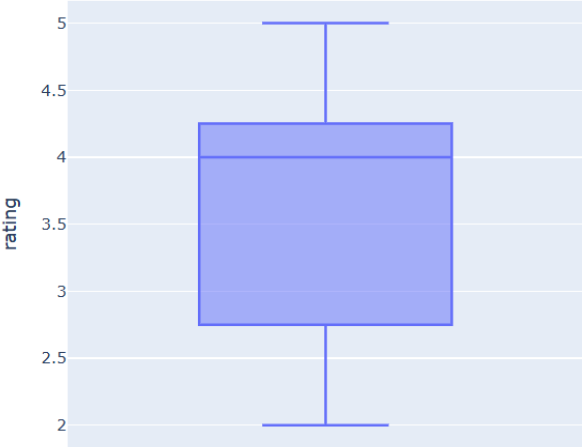
Delivery Time



Top Restaurants



Ratings



## 2. Bank Loan Approval Analysis

```
data = {
    'income': [50000, 60000, np.nan, 45000, 70000],
    'loan_amount': [200000, 250000, 180000, 150000, 300000],
    'credit_score': [700, 750, 720, 680, 800],
    'approval_status': ['Approved', 'Rejected', 'Approved',
                        'Rejected', 'Approved']
}

df = pd.DataFrame(data)
selected = df[['income', 'loan_amount', 'credit_score',
               'approval_status']]

from sklearn.impute import SimpleImputer from sklearn.preprocessing
import StandardScaler

imputer = SimpleImputer(strategy='mean')
selected['income'] = imputer.fit_transform(selected[['income']])

scaler = StandardScaler()

selected['loan_amount'] =
scaler.fit_transform(selected[['loan_amount']])

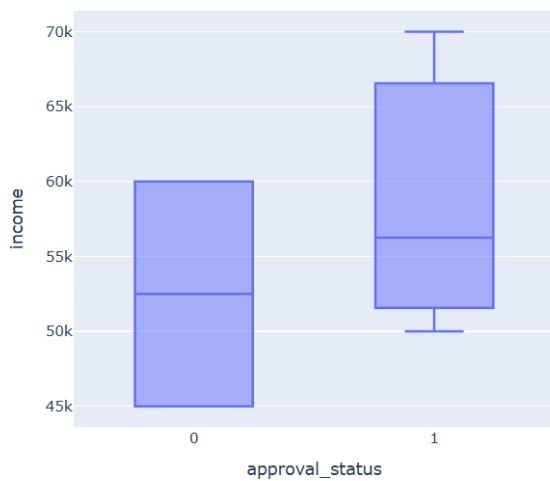
selected['approval_status'] =
selected['approval_status'].map({'Approved':1, 'Rejected':0}) print(selected)
   income  loan_amount  credit_score  approval_status  0  50000.0  -0.301084
700      1
1   60000.0   0.639803         750             0
2   56250.0  -0.677439         720             1
3   45000.0  -1.241971         680             0
4   70000.0   1.580691         800             1
fig1 = px.box(selected, x='approval_status', y='income', title="Income vs Approval")
```

```
fig1.show()
```

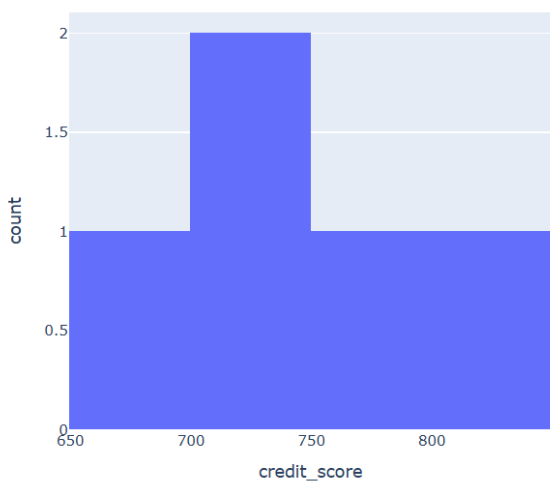
```
fig2 = px.histogram(selected, x='credit_score', title="Credit  
Score") fig2.show()
```

```
fig3 = px.scatter(selected, x='loan_amount', y='income',  
color='approval_status', title="Loan vs Income") fig3.show()
```

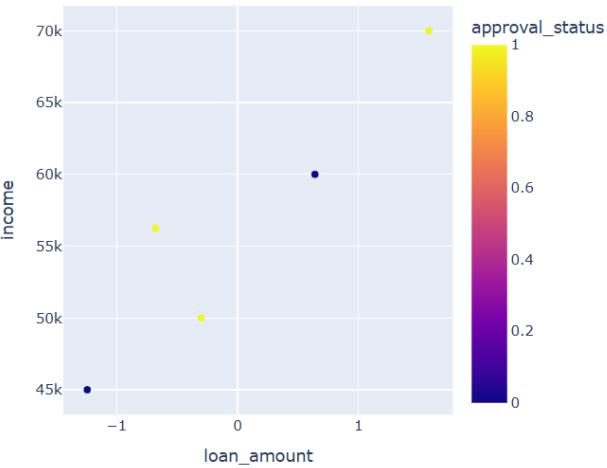
Income vs Approval



Credit Score



Loan vs Income



### 3.Smart Energy Consumption Monitoring

```
data = {
    'date': ['2026-01-01','2026-01-01','2026-01-02','2026-01-02'],
    'appliance': ['AC','Fan','AC','Light'],
    'power': [5, 1, 6, 0.5]
}

df = pd.DataFrame(data)

df['date'] = pd.to_datetime(df['date'])

grouped = df.groupby(['date','appliance'])['power'].sum()

print(grouped)

date    appliance
2026-01-01 AC      5.0
           Fan      1.0
2026-01-02 AC      6.0
           Light    0.5
Name: power, dtype: float64 df.loc[2, 'power'] =

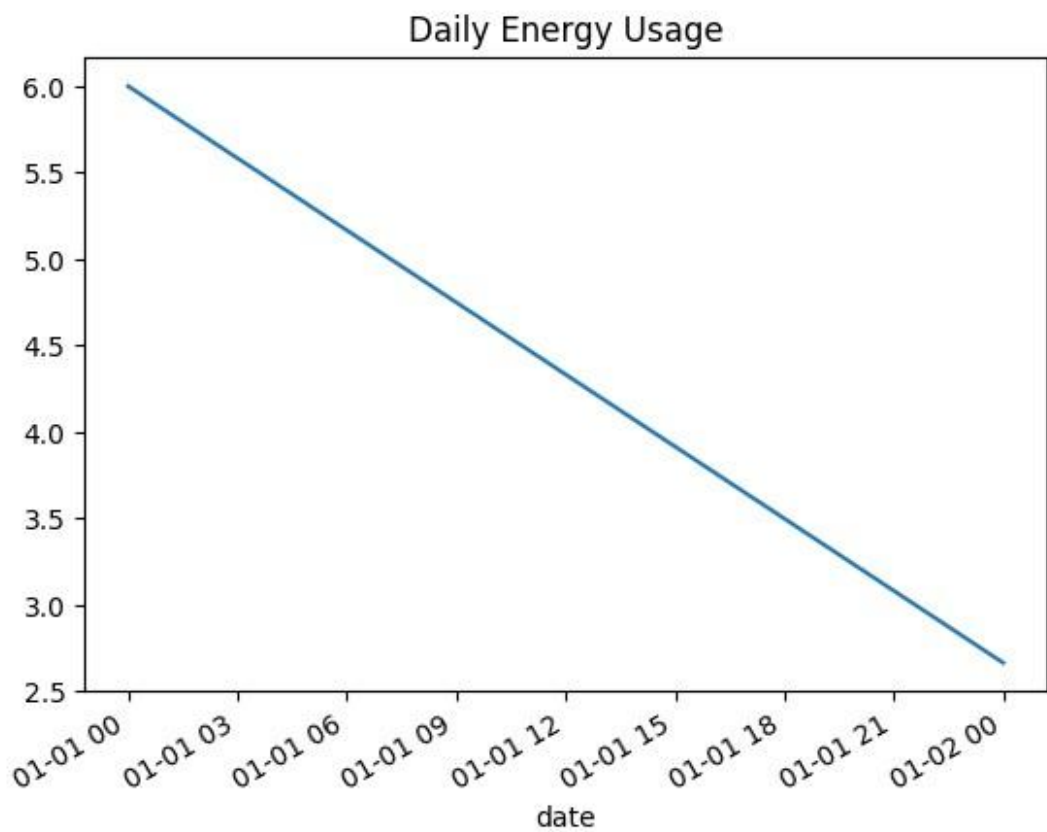
np.nan

df['power'] = df['power'].fillna(df['power'].mean()) df = df[df['power'] < 10]

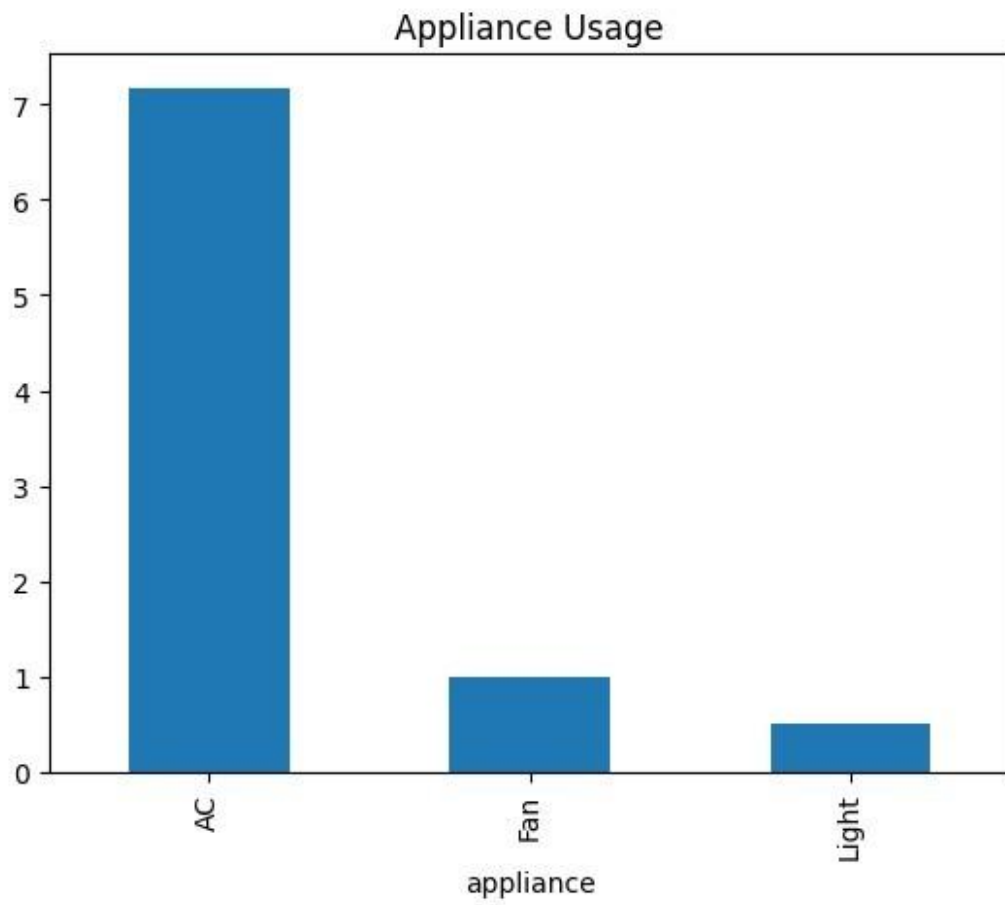
total = df.groupby('date')['power'].sum() print(total)

date
2026-01-01    6.000000
2026-01-02    2.666667
Name: power, dtype: float64
```

[illegible]







#### 4. Online Learning Platform Performance Analysis

```
df['quiz_score'] = df['quiz_score'].fillna(df['quiz_score'].mean())

df['performance'] = df['quiz_score'].apply(
    lambda x: 'High' if x > 80 else 'Medium' if x > 60 else 'Low'
) print(df)
```

|   | student | quiz_score | completion | rank | performance | 0 | A         |
|---|---------|------------|------------|------|-------------|---|-----------|
|   | 70      | 2.0        | Medium     |      |             |   | 80.000000 |
| 1 | B       | 60.000000  | 50         | 3.0  | Low         |   |           |
| 2 | C       | 90.000000  | 95         | 1.0  | High        |   |           |
| 3 | D       | 76.666667  | 60         | NaN  | Medium      |   |           |

```
import seaborn as sns import
matplotlib.pyplot as plt

sns.histplot(df['quiz_score']) plt.title("Quiz Score Distribution")
plt.show()

sns.barplot(x='student', y='completion', data=df) plt.title("Student
Engagement") plt.show()
```

```
data = {
    'student': ['A','B','C','D'],
    'quiz_score': [80, 60, 90, None],
    'completion': [70, 50, 95, 60]
}

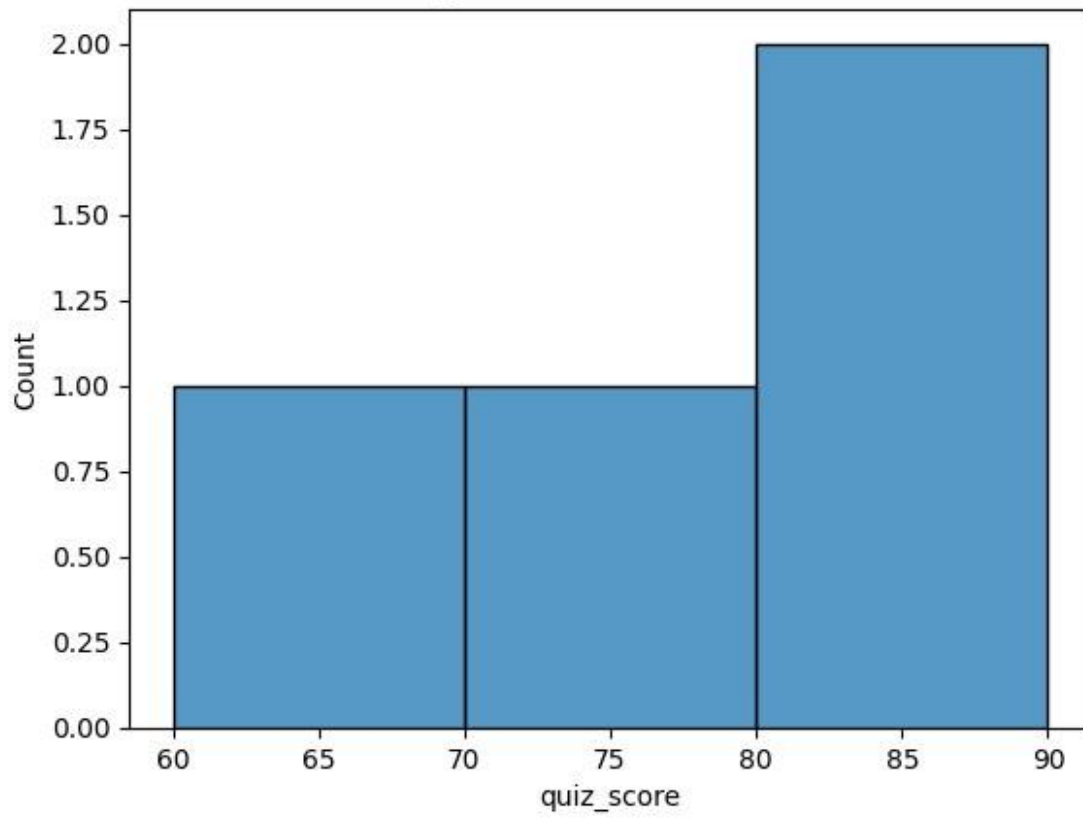
df = pd.DataFrame(data)

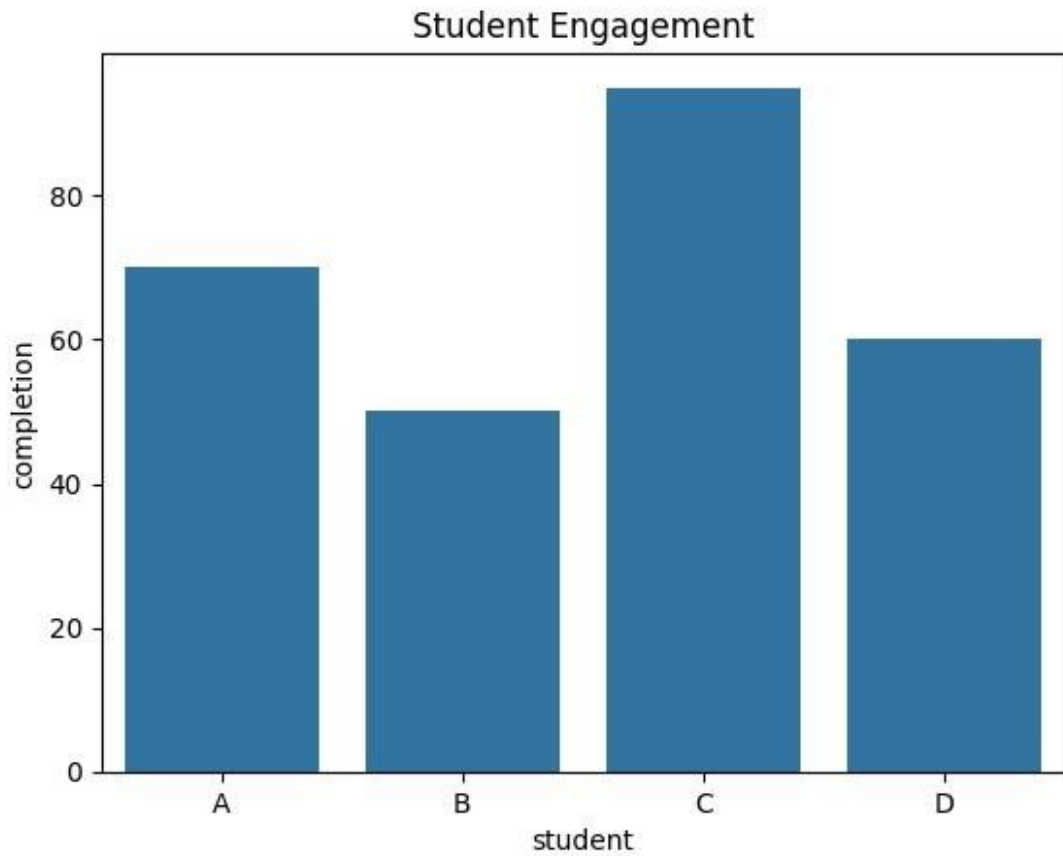
df_sorted = df.sort_values(by='quiz_score', ascending=False) filtered = df[df['completion'] > 60]

df['rank'] = df['quiz_score'].rank(ascending=False) print(df)
```

|   | student | quiz_score | completion | rank | 0 | A    |
|---|---------|------------|------------|------|---|------|
|   | 70      | 2.0        |            |      |   | 80.0 |
| 1 | B       | 60.0       | 50         | 3.0  |   |      |
| 2 | C       | 90.0       | 95         | 1.0  |   |      |
| 3 | D       | NaN        | 60         | NaN  |   |      |

Quiz Score Distribution





#### 5.Public Transport Passenger Data Analysis

```
data = {  
    'route': ['R1','R1','R2','R2'],  
    'stop': ['S1','S2','S1','S2'],  
    'time': ['Morning','Evening','Morning','Evening'],  
    'passengers': [50, 70, 80, 100]  
}  
  
df = pd.DataFrame(data)  
  
route_data = df.groupby('route')['passengers'].sum() time_data =  
df.groupby('time')['passengers'].sum()  
  
print(route_data) print(time_data)  
  
route  
R1    120  
R2    180  
Name: passengers, dtype: int64  
time  
Evening    170
```

Morning 130

Name: passengers, dtype: int64

```
df.loc[1, 'route'] = None df['route'] = df['route'].fillna('Unknown') df['passengers'] =
```

```
df['passengers'].fillna(df['passengers'].mean())
```

```
df = df.drop_duplicates() print(df)
```

|    | route | stop    | time | passengers | 0 | R1 | S1 | Morning |
|----|-------|---------|------|------------|---|----|----|---------|
| 50 |       |         |      |            |   |    |    |         |
| 1  |       | Unknown | S2   | Evening    |   |    |    | 70      |
| 2  |       | R2      | S1   | Morning    |   |    |    | 80      |
| 3  |       | R2      | S2   | Evening    |   |    |    | 100     |

```
import matplotlib.pyplot as plt
```

```
time_data.plot(kind='bar', title="Peak Travel Hours") plt.show()
```

```
route_data.plot(kind='bar', title="Busiest Routes") plt.show()
```

